1.1)

Hypothesis	θ	$P_0(\theta)$
Fair	0.5	0.50
Slightly biased	0.7	0.30
very Biased	0.9	0.20

Before the coin is tossed

Which hypothesis do you think is more likely?

The fair hypothesis is 50%

coin toss: heads

Bias hyphothesis	$P(\theta \mid H)$	$P_1(\theta)$	$\alpha P(H \mid \theta)$ Score	$P(H \mid \theta)$	Δ
None	.50	.50	.25	.44	-.06
slight	.70	.30	.21	.37	+.07
very	.90	.20	.11	.19	-.19

The outcome of heads is expected, so it doesn't move the posterior much

coin toss: heads

Bias hypothesis	$P(\theta \mid H)$	$P_2(\theta)$	$\alpha P(H \mid \theta)$	$P(H \mid \theta)$	Δ
None	.50	.44	.22	.34	-.10
Slight	.70	.37	.26	.40	+.03
very	.90	.19	.17	.26	+.07

$$P_n(\theta) = P_{n-1}(H|\theta)$$

$$P_3(\theta) - P_0(\theta)$$

H	$P_3(\theta)$	$P_0(\theta)$	$P_3(\theta) - P_0(\theta)$
none	.34	.50	−.16
Slight	.40	.30	+.10
very	.26	.20	+.06

Intuition check

Scientist A believes a coin is almost certainly fair
Scientist B believes it's probably heavily biased

They both watch the first two flips

Question:

Will they necessisarily reach the same conclusion after only a few observations?

Why or why not?

Their beliefs will converge over successive tests. Observations that are the most surprising will cause the biggest change in beliefs, so the one whose beliefs are the most out of line with reality will update the fastest, while the one that is closer to reality will not change nearly as much.